

The Utility Problem

With new trends and challenges in the energy sector, utilities need to be smart about every step they take. This is not true only on a macro level, but also on a micro level. Decisions on questions such as “Where should we deploy smart meters/IoT sensors next?”, “Which grid area needs inspection first?”, “What part of the grid needs OPEX to improve the power quality?”, or “Which parts of the grid are experiencing congestion?” are important to answer. The decisions on everyday questions like these have a significant impact on the effectiveness of grid operations. Proper and systematic prioritization is a key technique for many analysts addressing this topic.

For utilities, it is important that decision-making keeps pace with the trends and uses innovative techniques for data analytics as well. One of the innovative data analytics techniques in the electrical distribution utility space for prioritization is the creation and application of weighted analysis models for grid segments or grid elements.

The building and application of weighted models are in contrast to single-factor analytics. Single-factor analyses are often conducted through the lens of a single indicator - factor. This approach is fine, but it doesn't allow for a combination of multiple factors. Furthermore, different factors indicate different effects, thus it is important to include them in the analysis.

For these reasons, it is important to have the ability to systematically create and apply analysis models to distribution grid segments and individual elements. The results of such analysis allow prioritization for the next best action in the grid location where the action has the biggest impact.

Another important element is the correct visualisation of this data. By displaying the data not only in tables but also on a map, a more efficient evaluation can be achieved as the advantages of both methods are used.

The Data Solutions and Methods

The Awesense Platform allows for the division of the grid using the Segmentation tool into different-sized segments. Each segment then is a containerized part of the grid with the subset of the grid elements. Once the segments are created, the multi-factor weighted analysis model can be applied to the segments. This is done using the Segment Analysis tool. Before the application of the analysis, it is important to create the model which will be applied.

The process of building analysis models is as follows:

1. Identify a data source from which a single model factor will be created. The exact type of data depends on the role and use case of the desired model. For example, for non-technical loss factors, data can be a number of smart meters tamper flags. For transformer asset management, it can be the average loading of the transformer, etc. Simply, the data factor should be some kind of indication of the things we want to

prioritize within the model. All data are available in the Awesense Energy Data Model layer.

2. Create a factor by choosing the data source. Assign the calculation method, which will be executed over the segment (group of elements) or a single element.
3. Create a model by the combination of the factors. Each of the factors in the model can be assigned its own weight to influence their overall contribution to the final score. All these models can be created very easily and without programming knowledge by selecting from predefined functions or drop-down menus.

After the model is created, it can be applied to a segment group of elements or individual elements. These segments and/or elements then obtain a certain score based on the calculation from the model. The resulting values can be viewed not only on a map using all the functionality provided by the Awesense frontend solution of TGI, but also in tables directly in TGI, where flexible graphs can be created to complement the analysis already created (again, without the need for programming knowledge).

This functionality has almost unlimited use. For example, it can be used for penetration of electric vehicle chargers, distributed generation, energy balance, analysis of any elements in the grid and many more. This tool offers a great way of prioritizing grid areas for the next best action regardless of the use case.

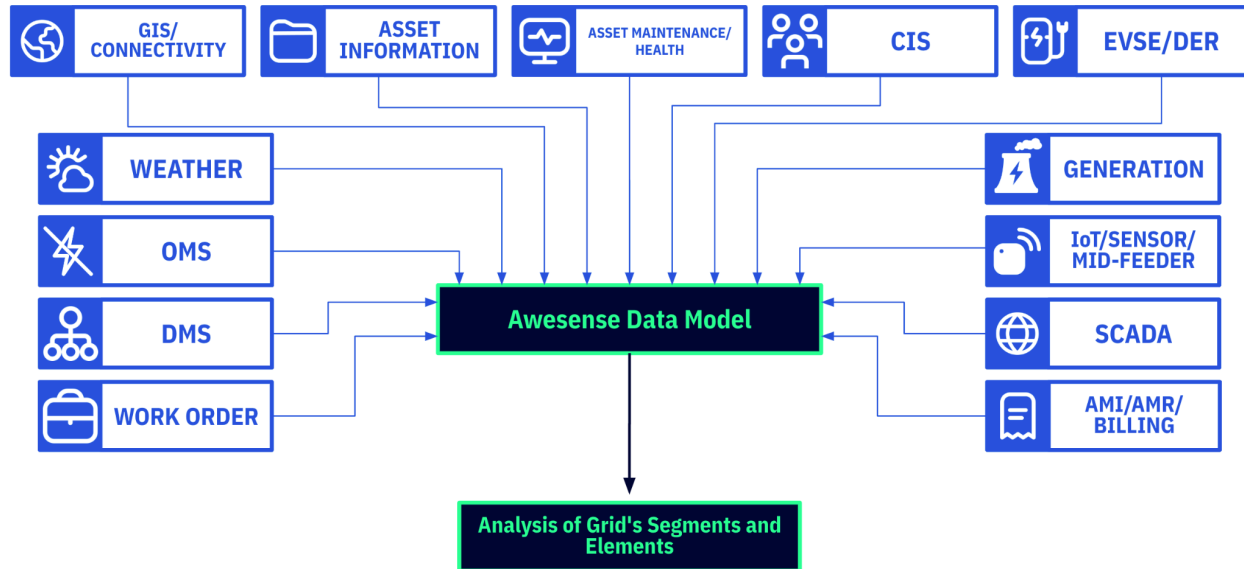
Analysis of Grid Segments and Elements

[UC23] - Awesense Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case. This use case can work with any type of the data which are supported by the Awesense Platform.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Segment Analyzer functionality. Consult [instructions](#) about how to access this use case in TGI. If you do not have access to a server where the use case functionality is available, contact [Awesense](#) to see a demo of it.